THANH VU

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EDUCATION

Franklin & Marshall College, Lancaster, PA *May 2026*

B.A. in Computer Science, Magna Cum Laude | GPA 3.72/4.0 | Honors List (5 semesters); Dean's List (Spring 2022)

- Computer Science: Algorithms, Artificial Intelligence, Machine Learning, Information Theory, Theory of Computer Science, Computational Discrete Math, Computer Organization
- Math, Statistics & Logic: Symbolic Logic, Intro to Data Science, Statistical Modeling, Probability & Statistics, Applied Regression, Calculus II–III

RESEARCH INTERESTS

Machine learning, probabilistic reasoning, and applied statistics for noisy real-world data, plus reliable data systems that make AI reproducible. Aim: data engineering now, then deeper study of deep learning and data preparation in the master's.

RESEARCH & APPLIED PROJECTS

Grocery Transaction Data Warehouse | SQL Server, T-SQL *Jul–Aug 2025*

- Built a Medallion warehouse over 131M+ transaction lines with stored-procedure ETL, a star-schema Gold layer, and 21 data-quality checks; tuning cut processing time 25%.

Sales Planning Analysis | Power BI, DAX, SQL Server *Aug 2025*

- Modeled 58K+ transactions in a galaxy schema and built Power BI reports showing plan completion of 89%–98% across districts.

Optimizing Decision Trees with Information Theory | Information Theory course project, co-written *Fall 2024*

- Co-wrote an 18-page paper and slides with a classmate; wrote the sections on entropy and information gain, feature selection, and pre- and post-pruning, and checked each other's sections.

Machine Learning Coursework (CPS 360) | Python, NumPy *Spring 2026*

- Wrote k-means and linear/logistic gradient descent from scratch.

EXPERIENCE

Data Analyst Fellow | EDC Lancaster County, Center for Regional Analysis (via The Shalom Project) *Aug 2026 – Present*

- Replaced manual Excel downloads with an automated pipeline, validation checks and a star-schema Power BI model (25+ BLS, BEA, Census, USPS series; 20 years; daily refresh), saving 15+ hours/month.
- Built a Google Apps Script web app with a Gemini AI assistant that flags anomalies, from requirements gathered by listening to a business analyst and the director, so 6+ non-technical colleagues update and explore the data without code.

Data Science Intern | KCC Capital Partners, Sheridan, WY *Jan – May 2026*

- Entered and cleaned data on 100+ mining companies in Excel and enriched contacts with Seamless.AI, screened it with the firm's built-in AI tools, and reviewed a 100+ buyer shortlist; automated weekly Excel reports, saving 5+ hrs/week.

Data Engineer Intern | KPIM Joint Stock Company, Hanoi, Vietnam *Jan – Apr 2021*

- Built Python and SQL Server ETL pipelines that automated legacy client data workflows, cutting processing time 30%.

LEADERSHIP

Technology Committee | Diplomatic Congress, Franklin & Marshall College *Aug 2021 – Aug 2022*

- Led IT policy work and centralized license management across 15+ departments, cutting classroom tech costs 18%.

SKILLS & CERTIFICATIONS

- **Programming:** Python (pandas, NumPy, scikit-learn, PySpark), SQL (T-SQL, Spark SQL), R, C++, Java
- **ML & Data:** regression, logistic regression, k-means clustering, model evaluation; Bayesian networks and state-space search (AI course); Microsoft Fabric, SQL Server, Power BI, ETL, data modeling, Git
- **Certifications:** Microsoft Fabric Data Engineer Associate (DP-700), Oct 2026; HackerRank SQL (Advanced), Sep 2025